Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int n, i, j;
    scanf("%d", &n);
    int a[n];

    for(i=0;i<n;i++) scanf("%d",&a[i]);

    for(i=0;i<n;i++)
        for(j=i+1;j<n;j++)
            if(a[i]==a[j]) {
                printf("%d",a[i]);
                return 0;
            }
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

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int main() {
    int n, i, j;
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    int a[n];

    for(i=0; i<n; i++)
        scanf("%d", &a[i]);

    for(i=0; i<n; i++)
        for(j=i+1; j<n; j++)
            if(a[i] == a[j]) {
                printf("%d", a[i]);
                return 0;
            }

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}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
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Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int T,n,m,i,j;
    scanf("%d",&T);

    while(T--) {
        scanf("%d",&n);
        int a[n];
        for(i=0;i<n;i++) scanf("%d",&a[i]);

        scanf("%d",&m);
        int b[m];
        for(i=0;i<m;i++) scanf("%d",&b[i]);

        i=j=0;
        while(i<n && j<m) {
            if(a[i]==b[j]) printf("%d ",a[i]),i++,j++;
            else if(a[i]<b[j]) i++;
            else j++;
        }
        printf("\n");
```

```
        printf("\n");
    }
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

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2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main(){
    int T,n,m,i,j;
    scanf("%d",&T);
    while(T--){
        scanf("%d",&n);
        int a[n];
        for(i=0;i<n;i++) scanf("%d",&a[i]);
        scanf("%d",&m);
        int b[m];
        for(i=0;i<m;i++) scanf("%d",&b[i]);
        i=j=0;
        while(i<n&&j<m)
            if(a[i]==b[j]) printf("%d ",a[i]),i++,j++;
            else if(a[i]<b[j]) i++;
            else j++;
        printf("\n");
    }
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	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

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Back to Course

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main() {
    int n,k,i,j;
    scanf("%d",&n);
    int a[n];
    for(i=0;i<n;i++) scanf("%d",&a[i]);
    scanf("%d",&k);

    i=0; j=1;
    while(j<n) {
        if(a[j]-a[i]==k) { printf("1"); return 0; }
        if(a[j]-a[i]<k) j++;
        else i++;
        if(i==j) j++;
    }
    printf("0");
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔

	Input	Expected	Got	
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

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1 - If pair exists

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Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main(){
    int n,k,i,j=1;
    scanf("%d",&n);
    int a[n];
    for(i=0;i<n;i++)
    scanf("%d",&a[i]);
    scanf("%d",&k);

    i=0;
    while(j<n){
        if(a[j]-a[i]==k){printf("1");return 0;

        }
        if(a[j]-a[i]<k) j++;
        else i++;
        if(i==j) j++;
    }
    printf("0");
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔

	Input	Expected	Got	
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

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